Sriram Balasubramanian

College Park, MD   sriramb@umd.edu   www.sriram.live   sriramb1998   Google Scholar

Education

University of Maryland, College Park — *August 2023 – December 2026 (Expected)*
PhD in Computer Science (advised by Prof. Soheil Feizi) — GPA: 4.0

University of Maryland, College Park — *August 2021 – May 2023*
MS in Computer Science (advised by Prof. Soheil Feizi) — GPA: 4.0

Indian Institute of Technology, Bombay — *August 2016 – May 2020*
B. Tech (Hons) in Computer Science & Engineering — GPA: 9.56/10.0
(advised by Prof. Sunita Sarawagi)

Work Experience

Research Fellow (Incoming) — *Berkeley, CA*
FAR.AI — *May 2026 – August 2026*

Research Intern — *San Jose, CA*
Adobe (Document Intelligence Lab) — *May 2025 – August 2025*

- Created a recipe to curate data and *post-train LLMs* using SFT and RL based algorithms to generate *citations* to answers by utilizing a decomposition based strategy. Preprint available here.
- Designed a new prompting strategy and a prompt optimization pipeline to improve attribution methods in Acrobat AI assistant by up to 10%.

AI Researcher — *College Park, MD*
RelAI — *June 2023 - August 2023*
University of Maryland, College Park — *June 2024 - August 2024*

- Developed and designed multiple applications to enhance vision model reliability for RelAI
- Involved in the development of RelAI, contributing from the inception stage through planning, execution, and deployment phases.

Research Intern — *Washington D.C.*
Comcast — *June 2022 - August 2022*

- Investigated the effectiveness of transfer learning in deep neural networks in the low resource regime (when the target domain has very limited data).
- Devised non-neural methods which could outperform both traditional collaborative filtering methods and neural networks in this regime.

Research Fellow — *Bangalore, India*
Microsoft Research — *August 2020 – August 2021*

- **Predicting e-mail arrivals and reads:** Built machine learning models to predict e-mail arrivals and reads from user type and history of arrivals/reads to improve cache hit rates.
- **Simulating network paths using ML:** Built machine learning models to simulate internet paths using static network traces

Intern — *Gurgaon, India*
Tower Research Capital — *May 2019 – August 2019*

- Developed tools to analyze trading and market logs, and compute and display metrics for various trading strategies and securities.

Publications and Preprints

Retrieval is Enough: Training-Free Interpretability with a Tool-Using Agent
S Balasubramanian, S Feizi
Under review at NeurIPS, (link here), 2026

Retrieval is Enough: Training-Free Interpretability with a Tool-Using Agent
HK Tan*, S Kabir, S Agrawal, S Chereddy, **S Balasubramanian**

Mentoring at SPAR, In Progress (link here), 2026

Decomposition-Enhanced Training for Post-Hoc Attributions In Language Models
S Balasubramanian, S Basu, K Goswami, R Rossi, V Manjunatha, R Santhosh, R Zhang, S Feizi, N Lipka
European Chapter of the Association for Computational Linguistics, Main, 2026

A Closer Look at Bias and Chain-of-Thought Faithfulness of Large (Vision) Language Models
S Balasubramanian, S Basu, S Feizi
Empirical Methods in Natural Language Processing (EMNLP), Findings, 2025

Decomposing and Interpreting Image Representations via Text in ViTs Beyond CLIP
S Balasubramanian, S Basu, S Feizi
Spotlight at *Mechanistic Interpretability Workshop, ICML*, 2024
Advances in Neural Information Processing Systems (NeurIPS), 2024

Exploring Geometry of Blind Spots in Vision Models
S Balasubramanian*, G Sriramanan*, VS Sadasivan, S Feizi
Spotlight at *Advances in Neural Information Processing Systems (NeurIPS)*, 2023

Towards Improved Input Masking for Convolutional Neural Networks
S Balasubramanian, S Feizi
IEEE/CVF International Conference on Computer Vision (ICCV), 2023

What's in a Name? Are BERT Named Entity Representations just as Good for any other Name?
S Balasubramanian*, N Jain*, G Jindal*, A Awasthi, S Sarawagi
Rep4NLP Workshop @ Annual Meeting of the Association of Computational Linguistics (ACL), 2020

Can AI-Generated Text be Reliably Detected? Stress Testing AI Text Detectors Under Various Attacks
VS Sadasivan, A Kumar, **S Balasubramanian**, W Wang, S Feizi
Transactions on Machine Learning Research (TMLR), 2025
Media coverage at Washington Post , Wired , TechSpot , New Scientist

Rethinking Copyright Infringements in the Era of Text-to-Image Generative Models
M Moayeri, **S Balasubramanian**, S Basu, P Kattakinda, A Chegini, R Brauneis, S Feizi
International Conference on Learning Representations (ICLR), 2025

Gaming Tool Preferences in Agentic LLMs
K Faghih, W Wang, Y Cheng, S Bharti, G Sriramanan, **S Balasubramanian**, P Hosseini, S Feizi
Empirical Methods in Natural Language Processing (EMNLP), Main Conference, 2025

Simulating Network Paths with Recurrent Buffering Units
D Anshumaan*, **S Balasubramanian***, S Tiwari, N Natarajan, S Sellamanickam, VN Padmanabhan
AAAI Conference on Artificial Intelligence (AAAI), 2023

Hop, Skip, and Overthink: Diagnosing Why Reasoning Models Fumble during Multi-Hop Analysis
A Yadav, I Nalawade, S Pillarichety, Y Babu, R Ghosh, S Basu, W Zhao, W Zhao, A Nasaeh, **S Balasubramanian**, S Srinivasan
arXiv preprint arXiv:2508.04699, 2025

Seeing What's Not There: Spurious Correlation in Multimodal LLMs
P Hosseini, S Nawathe, M Moayeri, **S Balasubramanian**, S Feizi
arXiv preprint arXiv:2503.08884, 2025

A Survey on Mechanistic Interpretability for Multi-Modal Foundation Models
Z Lin, S Basu, M Beigi, V Manjunatha, RA Rossi, Z Wang, Y Zhou, Y Zhou, **S Balasubramanian**, A Zarei, K Rezaei, Y Shen, B Menglong Yao, Z Xu, Q Liu, Y Zhang, Y Sun, S Liu, L Shen, H Li, S Feizi, L Huang
arXiv preprint arXiv:2502.17516, 2025

Services and Teaching

- **SPAR** **Mentor**: Mentoring 4 researchers (across multiple timezones) on projects related to utilizing circuits for solving AI safety problems
- Reviewer for prominent machine learning conferences such as ICML 2024, NeurIPS 2024 (Top Reviewer), ICLR 2025, NeurIPS 2025 (Top Reviewer), ICLR 2026, NeurIPS 2026.
- Introduced high-school students to AI as an instructor as part of TRAILS AI Summer Camp
- Teaching Assistant for Programming Handheld Systems (CMSC 436), Probability and Statistics (STAT 400) at UMD College Park; and Data Interpretation and Analysis (CS 215) and Electricity and Magnetism (PH 108) at IIT Bombay.

Awards and Honors

- Awarded Dean's Fellowship at the University of Maryland for the first year of PhD [2023]
- Awarded Institute Academic Prize for exceptional academic performance in IIT Bombay [2017]
- Ranked **2nd** in the institute out of about 900 students in the first year at IIT Bombay [2017]
- Ranked **4th** in JEE Mains out of 1.2 million candidates all over India [2016]
- Ranked **2nd** in the Maharashtra State Board Examinations (12th grade) [2016]